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WEEK :-1

DAY:-2

DATE:-06.09.2026

Literature Review: Domain + ML

1. Domain Understanding

Battery State of Charge (SOC) is a key state variable for battery-management systems. Direct measurement of SOC is difficult because it depends on battery chemistry, current, voltage, temperature, operating conditions and battery history. Machine-learning approaches are attractive because they can learn nonlinear relationships from measured data.

Remaining Charging Time (RCT) is also a useful prediction target for EV charging applications. Charging behaviour is not constant throughout a complete charge, so voltage, current, temperature, SOC and charging phase can influence the remaining time.

2. Literature Review Summary

- A review of machine learning state-of-charge and state-of-health estimation algorithms for lithium-ion batteries(2023):- Reviewed ML-based SOC

estimation methods and highlighted the importance of voltage, current and temperature as battery features.

- Deep Learning Framework for Lithium-ion Battery State of Charge Estimation: Recent Advances and Future Perspectives(2023):- Reviewed deep-learning approaches such as RNN and CNN can be used for accurate lithium-ion battery SOC estimation.
- State-of-Charge Estimation of a Li-Ion Battery using Deep Forward Neural Networks (2020): Applied a Deep Forward Neural Network for lithium-ion battery SOC estimation and demonstrated its potential for accurate prediction.
- Electric Vehicle Battery Remaining Charging Time Estimation Considering Charging Accuracy and Charging Profile Prediction (2022): Proposed an ML based approach for estimating the remaining EV battery charging time by considering charging profiles and conditions.
- A Comprehensive Review on the State of Charge Estimation for Lithium-ion Battery Based on Neural Network (2022): Reviewed neural-network approaches for SOC estimation and compared various deep-learning methods.
- A Method for State-of-Charge Estimation of Lithium-Ion Batteries Based on PSO-LSTM(2021): Proposed a PSO-LSTM model for SOC estimation and showed improved accuracy and robustness
- Overall Finding:

Hence, the studies show that ML/DL can effectively estimate SOC, while charging time prediction requires consideration of changing battery and charging conditions.

3. Candidate Methods

- Linear Regression:- Linear Regression can be used as a simple baseline model.

Advantages:

- Simple
- Fast
- Easy to interpret

Limitation:

- May not capture complex nonlinear battery behaviour.

- Decision Tree Regression:- Decision Tree Regression can model nonlinear relationships.

Advantages:

- Easy to understand

- Can model nonlinear relationships

Limitation:

- Can overfit the training data.

- Random Forest Regression:- Random Forest combines multiple decision trees.

Advantages:

- Handles nonlinear relationships
- Generally robust
- Works well with structured/tabular data

Limitation:

- More computationally expensive than a single tree
- Less interpretable than Linear Regression.

- Gradient Boosting / XGBoost :- Boosting models build several weak models sequentially to improve prediction performance.

Advantages:

- Often performs well on structured datasets
- Can capture complex relationships

Limitation:

- Requires tuning
- Can over fit if not controlled.

4. Proposed Evaluation Metrics

For both SOC and RCT:

- MAE — Mean Absolute Error
- RMSE — Root Mean Squared Error
- R^2 — Coefficient of Determination
- MSE— Mean Squared Error

Metrics must be reported separately because SOC and RCT have different units and meanings.

5. Research Gap / Motivation

The project will focus on a reproducible comparison of baseline and advanced regression models for two related charging-stage outputs: State of charge and Remaining Charging Time. Particular attention will be given to data quality, leakage-safe splitting by charging session/cycle where possible, temperature effects and model generalization.

6. Candidate-Method

Primary method:-

1. Linear Regression
2. Decision Tree Regression
3. Random Forest Regression
4. Gradient Boosting / XGBoost

Possible Deep-Learning Approaches: ·

- Artificial Neural Network (ANN)